

MAT1100 Homework 2

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October 5 2026

Question 2

(a) *Proof.* We begin with a couple of cases.

$n = 1$: $S_1 = \{e\}$ and so its only normal subgroup is itself.

$n = 2$: $S_2 \cong \mathbb{Z}/2\mathbb{Z}$ whose only normal subgroups are itself and the trivial group.

$n = 3$: $|S_3| = 6$ so by Lagrange's theorem its subgroups must have orders 1, 2, 3 or 6 now none of its order 2 subgroups are normal as these are generated by a single transposition and hence do not contain the entire conjugacy class (2). However any subgroup of order 3 is A_3 which is isomorphic to $\mathbb{Z}/3\mathbb{Z}$ and is normal since it has index 2. $S_3/A_3 \cong \mathbb{Z}/2\mathbb{Z}$. S_3 also has the trivial group and itself as normal subgroups.

$n = 4$: The conjugacy classes of S_4 are (1), (2), (2, 2), (3), (4) with sizes 1, 6, 3, 8, 6 and a normal subgroup of S_4 must have order dividing 24 and be a union of conjugacy classes. Since the largest odd number dividing 24 is 3, if N is a normal subgroup of S_4 it must contain the conjugacy classes (1), (2, 2) and hence contain $K = \{e, (12)(34), (13)(24), (14)(23)\}$. Hence all such N were classified in question 1 as $\{e\}, K, A_4, S_4$ with quotients $S_4, S_3, \mathbb{Z}/2\mathbb{Z}, \{e\}$.

If $n \geq 5$, A_5 is simple which greatly reduces the problem. Suppose $N \triangleleft S_n$ then $N \cap A_n \triangleleft S_n$ and so $N \cap A_n \triangleleft A_n$. Since A_n is simple this implies that either $N \cap A_n = \{e\}$ or $N \supseteq A_n$.

Case 1. $N \supseteq A_n$. Then $[S_n : N] \leq [S_n : A_n] = 2$ and hence $N = A_n$ or $N = S_n$.

Case 2. $N \cap A_n = \{e\}$, now if $\sigma \in S_n$ then

$$\text{sgn}(\sigma^2) = \text{sgn}(\sigma)^2 = 1$$

hence $\sigma^2 \in A_n$. Therefore every element of N can have order at most 2.

Now since N is normal it is the union of conjugacy classes in S_n hence it is the union of conjugacy classes of the form $(2, \dots, 2)$ k times for $k \leq n/2$. First note that if N contains the conjugacy class for $k = 1$ then N contains all transpositions and is therefore all of S_n .

First suppose N contains the conjugacy class for $2 \leq k < n/2$. Then $2k < n$. Now pick $(i, j) \in S_n$ and pick $(i_1, j_1) \dots (i_{k-1}, j_{k-1})$ such that all of these transpositions are disjoint and do not contain either i or j . Then since $2k < n$ there exists $\ell \in \{1, \dots, n\}$ such that $\ell \neq i, j$ and ℓ is not contained in any of the above transpositions. Then

$$(i_1, j_1) \dots (i_{k-1}, j_{k-1})(k, j), (i_1, j_1) \dots (i_{k-1}, j_{k-1})(i, k) \in N$$

and their product is (i, j) therefore N contains all of the transpositions and is hence of all of S_n .

Next suppose N contains the conjugacy class for $k = n/2$. assuming n is even. Pick $i, j, r, s \in \{1, \dots, n\}$ all distinct as $n > 4$. We have $k \geq 0$, so let $(i_1 j_1), \dots, (i_{k-2} j_{k-2}) \in S_n$ be disjoint transpositions such that none of them contain any of i, j, r, s . Then

$$(i_1 j_1) \dots (i_{k-2} j_{k-2})(ir)(sj), (i_1 j_1) \dots (i_{k-2} j_{k-2})(is)(rj) \in N$$

and their product is

$$(ij)(rs)$$

hence N contains the conjugacy class $(2, 2)$ and by the above case this implies that $N = S_n$. Therefore $N = \{e\}$ if $N \cap A_n = \{e\}$ and the only normal subgroups of S_n are $\{e\}, A_n, S_n$ with quotients $S_n, \mathbb{Z}/2\mathbb{Z}, \{e\}$. $\square$

- (b) *Proof.* Let B be an abelian group. The zero map is always a homomorphism S_n to B so it is not stated. If $f : S_n \rightarrow B$ is a homomorphism then

$$S_n / \ker f \cong \text{im } f \leq B$$

hence it suffices to classify all subgroups of B are isomorphic to a quotient of S_n and for this it is clear that we need only consider the quotients of S_n which are abelian and for $n \geq 2$ the only non trivial such quotient is $S_n/A_n \cong \mathbb{Z}/2\mathbb{Z}$.

$n = 1$ The only homomorphism out of the trivial group into another group is the zero map.

$n = 2$ If $2 \mid |B|$ then by Cauchy's theorem B has an element x of order 2 then there is a map: $S_2 \rightarrow B$ given by $(12) \rightarrow x$ and such a map can be constructed for any element of B with order 2. These are clearly all of the maps from S_2 to B .

$n = 3$ If $2 \mid |B|$ then by Cauchy's theorem B has an element x of order 2 then there is a map: $S_3 \rightarrow B$ given by $\sigma \rightarrow x$ for $\sigma \in S_3 \setminus A_3$ and $\sigma \rightarrow e$ for $\sigma \in A_3$. Such a map can be constructed for any element $x \in B$ of order 2. Since S_3 has no abelian quotients other than $\{e\}$ and $\mathbb{Z}/2\mathbb{Z}$ these are all of the possible maps from S_3 into B (as S_3 is non abelian).

$n = 4$ If $2 \mid |B|$ then by Cauchy's theorem B has an element x of order 2 then there is a map: $S_4 \rightarrow B$ given by $\sigma \rightarrow x$ for $\sigma \in S_4 \setminus A_4$ and $\sigma \rightarrow e$ for $\sigma \in A_4$. Such a map can be constructed for any element $x \in B$ of order 2. Since S_4 has no abelian quotients other than $\{e\}$ and $\mathbb{Z}/2\mathbb{Z}$ these are all of the possible maps from S_4 into B (as S_4 is non abelian and $S_4/K \cong S_3$ is non abelian).

$n \geq 5$ If $2 \mid |B|$ then by Cauchy's theorem B has an element x of order 2 then there is a map: $S_n \rightarrow B$ given by $\sigma \rightarrow x$ for $\sigma \in S_n \setminus A_n$ and $\sigma \rightarrow e$ for $\sigma \in A_n$. Such a map can be constructed for any element $x \in B$ of order 2. Since S_n has no abelian quotients other than $\{e\}$ and $\mathbb{Z}/2\mathbb{Z}$ these are all of the possible maps from S_n into B as S_n is non abelian.

$\square$